A Coefficient-Aware Finite-Difference Benchmark for Solver Selection and CPU/GPU Stencil Scaling in Heat-Conduction Simulation

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Abstract

We present a verified finite-difference heat-conduction benchmark that tests whether coefficient summaries distinguish solver choices. Matched binary fields preserve contrast, material fraction, total variation and maximum log-gradient. At contrast 1000 and grid size 192, translating a two-island field by one node raises Jacobi-PCG iterations from 403 to 677 and reverses the setup-inclusive winner in all five repeats. Source symmetry explains the avoided slow modes; a classical-AMG audit shows that decision boundaries depend on the candidate set. All 2956 controlled extension solves pass an independent residual check. Smooth and aligned-interface tests verify the discretisation, while a paired calibration quantifies callback overhead separately from historical timings. CPU/CUDA results concern stencil execution with resident GPU data. The contribution is a reproducible audit of conditional solver costs, not a new selection algorithm.

1 Introduction and context

Variable-coefficient heat conduction couples discretisation accuracy to sparse-solver cost. Using SciPy, NumPy, Numba and PyAMG [1, 2, 3, 4], we audit when coefficient summaries fail to distinguish the fastest verified solve. Finite differences, conjugate gradients (CG) and algebraic multigrid (AMG) are established methods [5, 6, 7, 8]. Recent work addresses sparse multi-GPU efficiency and GPU PDE preconditioning [9, 10], communication-reduced AMG and anisotropic diffusion [11, 12], and reproducible benchmarking [13]. HPGMG-FV and AMG2023 already combine elliptic problems with performance measurements, including setup costs [14, 15]; we do not claim this combination as new.

Hierarchy quality can affect AMG convergence beyond scalar contrast [16]. Ginkgo separates linear-operator algorithms from architecture-specific backends [17], while stencil and CUDA studies expose sensitivity to memory traffic and kernel design [18, 19]. Three-precision AMG and adaptive-precision matrix-vector products further motivate explicit arithmetic settings [20, 21]; our experiments use float64. High-contrast island modes [22, 23], source-dependent Krylov convergence [24] and material-aware aggregation [25] supply the numerical context. Feature-based solver prediction [26], cheaper feature selection [27], multimodal solver/preconditioner selection [28] and online multiphysics solver choice [29] pursue selection policies. We instead test descriptor sufficiency with exact matched-field controls, without training a selector or benchmarking those policies.

Our contribution is the controlled connection between preserved coefficient invariants, forcing symmetry and measured setup-inclusive choices. Rigid translations preserve shape and connectivity as well as scalar descriptors; source and candidate-solver controls identify why a coefficient-only decision map cannot be reused uncritically.

2 Operator, verification and timing

On a uniform N-by-N grid of the unit square, with h = 1/(N − 1), positive conductivity k and homogeneous Dirichlet boundaries, the conservative operator is

$$-\nabla\cdot(k\nabla u) = f, \quad \Omega = (0,1)^2, \quad u|_{\partial\Omega} = 0, \qquad (Au)_i = h^{-2}\sum_{j\sim i} k_{ij}(u_i - u_j). \tag{1}$$

The sum runs over four neighbours; shared positive face weights make the interior matrix symmetric positive definite. Arithmetic faces define the historical nodal graph-diffusion benchmark. For physical interfaces, two equal half-links in series instead give the harmonic conductivity [30]:

$$k_H = \frac{2ab}{a+b}, \qquad k_A = \frac{a+b}{2}, \qquad k_A - k_H = \frac{(a-b)^2}{2(a+b)}. \tag{2}$$

The difference is O(h^2) for smooth positive k, but kA/kH = 25.5025 at contrast 100. This local conductance ratio is not a temperature-error estimate. Arithmetic stress tests therefore characterise the specified operator rather than validating composite-material flux; harmonic averaging resolves the aligned resistance model, but not a curved interface cutting cells. Both averages enter the matched-field audit.

With exact solution sin(πx)sin(πy), constant and smooth-variable tests give fitted L^2 orders 2.01 and 2.02 over N = 32–256. The aligned one-dimensional interface test gives harmonic flux errors of 5.11×10^{-15}–4.91×10^{-14}, versus arithmetic errors decreasing from 0.0633 to 0.00749. Full verification tables, coefficient definitions and implementation details appear in Supplement S1–S3. At N = 128 and contrast 100, inclusion and checkerboard condition estimates differ (6.43×10^5 versus 3.12×10^4), already showing that contrast alone is insufficient.

All runs use one Windows 11 workstation (Ryzen 9 7940HS, 16 GB RAM, RTX 4060 Laptop GPU). The principal candidates are CG, Jacobi-PCG (J) and default PyAMG 5.3.0 smoothed-aggregation PCG (SA). Selection minimises fresh preconditioner setup plus solve time; assembly and source construction are excluded, as is the cost of testing other candidates. Three protocols must be distinguished [31]. Historical comparisons use a unit source, rtol = 1×10^{-8}, per-iteration true-residual callbacks, fixed order and first-use PyAMG import inside setup; threads and power states were not recorded. The controlled extension uses warm imports, shuffled order, one BLAS thread, paired setup seeds and count-only callbacks, with rtol = 1×10^{-9}. Calibration holds these controlled execution settings fixed at rtol = 1×10^{-8} and changes only the callback within each pair. Zero initial guesses and zero absolute tolerances apply; extension acceptance requires an independent relative residual $\leq 1 \times 10^{-8}$. Timings and winner boundaries are compared within protocols, never pooled across them. Full defaults and iteration limits are in Supplement S2; environment versions are in S3.

3 Interface and callback sensitivity

The historical single-pass map spans 13 coefficient cases and three sizes: J wins at N = 64 and SA at N = 128, 256. The winner is constant across coefficient families at each size, so the map does not show a predictive benefit beyond resolution. Five-repeat checks retain the median winner in every repeat in 14/15 cells; constant N = 64 is a near tie (J median advantage 1.03×, CG wins 4/5). For constant k, Jacobi is scalar scaling and gives no spectral improvement. Table 1 retains all nine face-average comparisons: matching winners do not establish physical equivalence, timing stability or equal solution accuracy.

Calibration uses arithmetic faces and a unit source for constant, inclusion and checkerboard fields (contrast 100 for the latter two), three sizes, three methods, two callbacks and seven shuffled repeats (378 solves). Every solve passes the independent 1×10^{-8} check; all 189 paired solutions have identical hashes and iteration counts. Monitored/count-only solve-time ratios span 1.54–1.84 for CG, 1.58–1.77 for J and 1.03–1.06 for SA (Table 2). All nine median winners agree across callbacks, but constant N = 256 has only 3/7 count-only SA wins. Median optimality and majority voting differ. Several N = 128 winners differ from the historical map under both callbacks: callback removal alone cannot explain the change. Calibration neither measures cold-import overhead nor controls operating-system, power or thermal variation.

Table 1: All contrast-100 historical face checks: arithmetic/harmonic iterations and setup-inclusive winners (J = Jacobi-PCG; SA = smoothed-aggregation PCG). Single timing passes; no uncertainty estimate.

Case	N	CG	J	SA	Winner
Inclusion	64	796/725	129/129	10/10	J/J
Inclusion	128	1963/1891	264/263	12/11	SA/SA
Inclusion	256	4370/4326	535/535	12/12	SA/SA
Layered	64	1136/1120	164/181	9/9	J/J
Layered	128	2447/2442	325/367	10/10	SA/SA
Layered	256	5055/5054	657/715	10/10	SA/SA
Checkerboard	64	443/557	171/190	13/28	J/J
Checkerboard	128	1050/1308	350/388	17/32	SA/SA
Checkerboard	256	2268/2809	718/783	17/31	SA/SA

Table 2: All callback-calibration cells. Winners: method, median total ms and wins/7. Last columns: paired median monitored/count-only solve-time ratios.

Case	N	Monitored	Count-only	CG	J	SA
Constant	64	CG 5.64 (7/7)	CG 3.55 (5/7)	1.58	1.58	1.05
Constant	128	CG 31.22 (6/7)	CG 19.11 (7/7)	1.67	1.70	1.04
Constant	256	SA 141.18 (7/7)	SA 147.70 (3/7)	1.84	1.76	1.04
Inclusion	64	J 6.67 (7/7)	J 4.12 (7/7)	1.63	1.62	1.03
Inclusion	128	J 38.28 (7/7)	J 22.34 (7/7)	1.72	1.61	1.04
Inclusion	256	SA 156.23 (7/7)	SA 152.34 (6/7)	1.79	1.77	1.03
Checkerboard	64	J 8.69 (6/7)	J 5.63 (7/7)	1.54	1.59	1.04
Checkerboard	128	J 50.18 (5/7)	J 29.65 (7/7)	1.73	1.65	1.06
Checkerboard	256	SA 192.31 (7/7)	SA 180.24 (7/7)	1.79	1.67	1.06

4 Matched-field, forcing and solver-set audit

Two 2-by-8 motifs have 12 high cells and 20 interface edges, but one or two connected components (Figure 1). High/background conductivities are c/1. Dilation by s = N/16 and low-conductivity padding give $12s^2$ high nodes, mean $1 + (c - 1)12s^2/N^2$, total-variation proxy $(c - 1)20s/(N - 1)$, and maximum nodal log-gradient $\sqrt{2}(N - 1)\log(c)/2$. Total variation sums absolute nearest-neighbour differences and divides by N − 1; the log-gradient uses centred interior and one-sided boundary differences. For jumps it is a fixed-grid descriptor. These identities are checked after embedding. A one-node translation preserves them, component sizes and shape, while changing alignment with the source and boundary. Comparing different motifs alone does not isolate topology causally.

After a 576-solve exploratory screen, confirmation uses c = 1000, N = 96, 128, 192, centred/transposed/x-shifted configurations, both face averages and four unit-norm sources: constant, $\sin(2\pi x)\sin(\pi y)$, $\sin(\pi x)\sin(2\pi y)$ and fixed-seed Gaussian. J and SA each have five fresh paired setups. The 1440 confirmation solves, 640 solves from eight prespecified 3-by-6 motifs, and 132 translation/source-perturbation solves pass acceptance (maximum residual $<3 \times 10^{-9}$). The independent family balances four connected and four disconnected patterns; it is not a random material sample. The pilot retains 84 failed acceptance checks, excluded from successful timings.

Let R reflect interior nodes in x, D = diag(A), $S = D^{-1/2} A D^{-1/2}$, and $g = D^{-1/2}b$. When RA = AR and Rb = −b, the zero-start Krylov space generated by S and g is odd and orthogonal to x-even eigenvectors. The centred two-island field has two slow x-even Jacobi-scaled modes; computed squared source overlaps are $<1 \times 10^{-30}$ at N = 64, c = 10000 for both face averages. Joint transposition preserves J iterations, whereas x translation breaks reflection symmetry. This is an invariant-subspace explanation, not new convergence theory.

At N = 192 with harmonic faces and odd-x forcing, J changes from 403 iterations/72.2 ms to 677/125.1 ms after translation; SA changes from 17/101.0 ms to 16/97.5 ms. Each position retains its winner in all five repeats. A random source on the centred operator gives J 949 iterations/166.2 ms versus SA 15/94.3

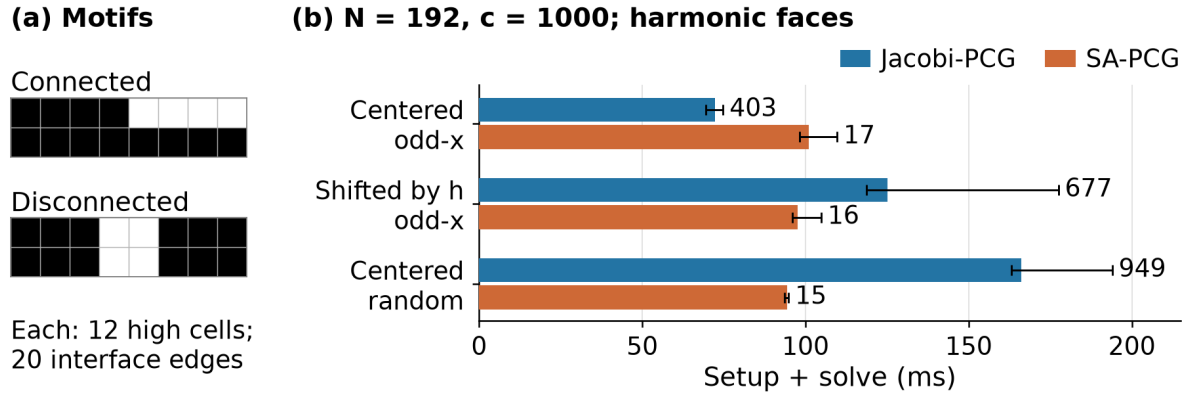


Figure 1: Matched motifs (left); two-island source/alignment sensitivity at N = 192, c = 1000, harmonic faces (right). Bars: median setup + solve time; whiskers: five-run range; labels: iterations.

ms. Adding only 1×10^{-8} of a constant component to the odd source raises J from 403 to 569 iterations. A y translation can preserve x reflection while changing the remaining spectrum; parity alone is not a runtime predictor.

Stable per-repeat winners occur in 124/144 confirmation cells and 55/64 independent-family cells; only 2/48 centred/shifted pairs reverse consistently in all five repeats. The result is a counterexample, not a prevalence estimate. Any deterministic choice restricted to the preserved invariants must select the same method for both positions. For measured median total time T(j,s), its least equally weighted excess cost over portfolio P is

$$E(P) = \min_{s \in P} \left[\frac{1}{2} \sum_{j=0}^{1} \frac{T(j,s)}{\min_{t \in P} T(j,t)} \right] - 1. \tag{3}$$

For this J/SA pair, E(P) = 14.2%. This empirical quantity is neither a statistical lower confidence bound nor a bound for other portfolios. A 240-solve screen and 504-solve interleaved confirmation add classical Ruge–Stüben AMG (strength threshold 0.25). Confirmation compares J, default SA and classical AMG at N = 128, 160, 192, for both positions, odd-x/random sources and two motifs: the original two-island field and a 3-by-6 anchor with rows 110011. All 744 solves pass acceptance. In seven-repeat confirmation, 17/24 cells retain the same winner across all seven repeats and 3/12 pairs reverse consistently (Table 3). For the original motif at N = 128 with odd-x forcing, centred J/classical-AMG times are 24.3/36.8 ms, versus 42.3/34.2 ms after translation; J iterations rise 266→450 while classical AMG takes 11 in both. E(P) = 11.91%. At N = 192, classical AMG wins both median comparisons and the floor is zero, with an unstable centred winner. The stronger baseline changes the decision boundary and cost consequence; these timings are not pooled with earlier stages.

5 Stencil scaling and implications

At N = 2048, variable-coefficient stencil times are 0.1813 s (NumPy), 0.005325 s (Numba serial) and 0.003725 s (Numba, four threads); more threads did not improve the largest case. Separate resident-data Jacobi updates give CUDA/CPU speedups near 1 at N = 512, then 1.52×, 3.23× and 3.51× at N = 1024, 2048, 4096. CPU uses four threads and CUDA 16-by-16 blocks. Host-clock timings include launch and final synchronisation of 20-update batches, excluding allocation, compilation and transfers. At least three batches and 0.2 s yield within-run averages, not independent replicates. These are kernel results, not end-to-end sparse-solver speedups. Full curves and byte models are in Supplement S3.5 and S2, respectively. A four-point linear fit predicts a negative CPU time at N = 512 and cannot establish a crossover threshold.

Table 3: All 12 interleaved translated pairs. RS = classical AMG-PCG; SA also competed. Stable: each position retains its winner in all seven repeats. E(P) uses medians even for unstable pairs; it is not a confidence bound.

N	Motif	Source	Centered	Shifted	Stable	$E(P)$ (%)
128	3x6	Odd-x	J	RS	Yes	12.42
128	3x6	Random	RS	RS	Yes	0.00
128	2x8	Odd-x	J	RS	Yes	11.91
128	2x8	Random	RS	RS	Yes	0.00
160	3x6	Odd-x	J	RS	Yes	9.69
160	3x6	Random	RS	RS	No	0.00
160	2x8	Odd-x	J	RS	No	4.86
160	2x8	Random	RS	RS	Yes	0.00
192	3x6	Odd-x	RS	RS	No	0.00
192	3x6	Random	RS	RS	Yes	0.00
192	2x8	Odd-x	RS	RS	No	0.00
192	2x8	Random	RS	RS	Yes	0.00

The audit shows why forcing, alignment and candidate solvers must be checked before reusing coefficient-based choices. Physical interpretation additionally requires a justified interface law, and timing interpretation requires the callback and setup protocol. Evidence remains limited to synthetic structured 2D problems and one workstation. Descriptive correlations have overlapping resampling intervals and resolution confounding (N = 256 uses an N = 128 spectral proxy); they do not establish descriptor rankings or population confidence. Neither production robustness nor universal solver thresholds follow from these measurements.

6 Reproducibility

Numerical evidence and scripts, including callback calibration, are archived at `https://doi.org/10.5281/zenodo.22874266` (v5). The accompanying supplement retains complete tables, plots, protocols and failures; the present compression changes presentation only. Tests cover flux energy, matched descriptors, reflection and transposition, with explicit CPU-only/CUDA-required modes. Timing reruns remain machine-state dependent.

Author contributions

Dong Liu led the conceptualization, formulation of the benchmark problems, numerical-method selection, and development of the finite-difference and CPU/GPU benchmarking workflow. He designed the numerical verification, computational-cost comparisons, and repeated-run experiments, and led the computational analysis and interpretation of the findings. He drafted and integrated the manuscript, with primary responsibility for verifying its scientific claims and defining their scope and limitations.

Yuanxin Wang contributed to the execution and verification of the solver-comparison experiments, comparing conjugate-gradient and preconditioned conjugate-gradient methods across different grid resolutions and conductivity distributions. He compiled iteration counts, preconditioner setup times, and solve times, and checked numerical outputs using the independent residual-verification procedure to support cost comparisons among runs meeting the prescribed accuracy criterion. He helped assess whether the fastest method remained consistent across repeated runs and how relative solver performance changed under different problem settings. He also contributed to comparative figures, interpretation of the corresponding results and their scope and limitations, and manuscript revision.

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